

Program : Diploma in Wood and Paper Technology	
Course Code : 5121	Course Title: Adhesive Technology
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge about the application of various adhesives in wood based industries.
- To impart knowledge about characteristics of glues and method of application on wood and wood products
- To familiarize the students manufacture of Adhesives

Course Prerequisites:

Topic	Course code	Course name	Semester
Polymerization, chemical reactions		Applied chemistry	1
Viscosity, surface tension		Applied physics I	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Classify the types of adhesives and explain their composition	15	Understanding
CO2	Discuss the physical chemical principles of gluing	17	Applying
CO3	Describe the manufacture and industrial applications of Thermosetting resins	17	Applying
CO4	Summarize the manufacture and industrial applications of Thermoplastic resins	9	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3		3			
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Classify the types of adhesives and explain their composition		
M1.01	Adhesives and their importance in wood industry	1	Understanding
M1.02	Classify adhesives	6	Understanding
M1.03	Explain the composition of adhesives and additives used	4	Applying
M1.04	Discuss the concepts of glue mixing	4	Applying
Contents: Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrans, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine, phenol, resorcinol..) and Thermoplastic resin (cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin.			
CO2	Discuss the physical chemical principles of gluing		
M2.01	Explain the rheology of glues	6	Understanding

M2.02	Discuss the glue spreading	5	Understanding
M2.03	Demonstrate the tests on glues and gluing defects	6	Applying
	Series Test – I	1	

Contents:

Physical chemical principles of gluing

Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening

Influence of - temperature & concentration on surface tension; P^H and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film

Pre-treatments of wood prior to gluing.

Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints.

Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde.

CO3	Describe the manufacture and industrial applications of Thermosetting resins		
M3.01	Classify synthetic resins and its types	2	Understanding
M3.02	Summarize the manufacture of thermosetting resins	8	Understanding
M3.03	Discuss the applications of thermosetting resins in wood industry	4	Applying
M3.04	Explain the working of Resin kettle	3	Understanding

Contents:

Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties & industrial applications of: Thermosetting resins

Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde (MF) resin, Urea melamine formaldehyde resin (UMF)

Phenolic resins - Phenol formaldehyde resin (PF) - resole and novolac

Resorcinol formaldehyde (RF), Phenol resorcinol formaldehyde (PRF), Resin kettle

CO4	Summarize the manufacture and industrial applications of Thermoplastic resins		
M4.01	Explain the manufacture of epoxy resins and thermoplastic resins	3	Understanding
M4.02	Compare the advantages of each adhesive over the other	3	Understanding

M4.03	Explain the properties and applications of thermoplastic resins	3	Understanding
	Series Test – II	1	
Contents: CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications			

Text / Reference:

T/R	Book Title/Author
T1	F.P Kollmann, Principles of wood science & Technology ,Springer
R1	Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, Adhesive & Chemical Products, New York, N.Y.
R2	Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical Division Adhesive & Chemical Products, New York, N.Y.
R3	Synthetic resins & their industrial applications – SBP – Board of consultants & Engineers New Delhi
R4	Lecture notes on plywood technology – IPIRI Bangalore1993
R5	Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill Inc., Rept., Nov. 22, 1965.
R6	Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive Guide, Furniture Desiv, & Manufacturing Magazine, January 1966.

Online Resources:

Sl.No	Website Link
1	https://files.eric.ed.gov/fulltext/ED031557.pdf
2	https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of-adhesive-technology.pdf
3	https://www.adhesives.org/adhesives-sealants/adhesives-sealants-overview/adhesive-technologies